

AI MEDICAL PRE-CONSULTATION SYSTEM



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INTRODUCTION

A bilingual web-based system that streamlines the pre-consultation process by collecting patient information and providing AI-powered preliminary diagnosis to physicians. The system enables patients to submit their medical data through traditional forms or voice interaction, while doctors receive AI-generated analysis before patient appointments, improving consultation quality and efficiency.

OBJECTIVE

To develop an intelligent pre-consultation system that automates patient data collection and provides preliminary medical analysis to physicians. The system aims to reduce consultation time, improve diagnostic accuracy, and enhance patient accessibility through voice-based interaction and bilingual support for both English and Arabic speakers.

METHODOLOGY

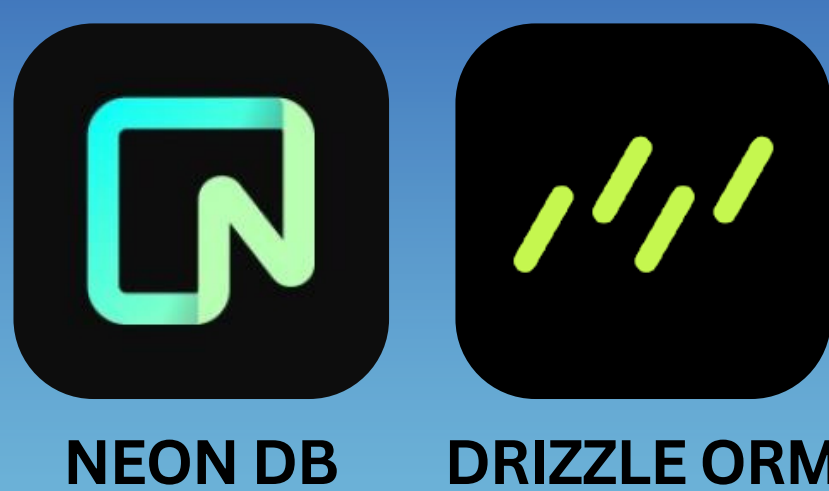
The system was developed using modern web technologies with a Next.js framework and PostgreSQL database. AI integration was implemented using OpenAI's language models for medical analysis and voice conversation capabilities. The architecture supports three user roles: patients who submit consultations, doctors who review reports, and administrators who manage the system.

IMPLEMENTATION TOOLS

DEVELOPMENT



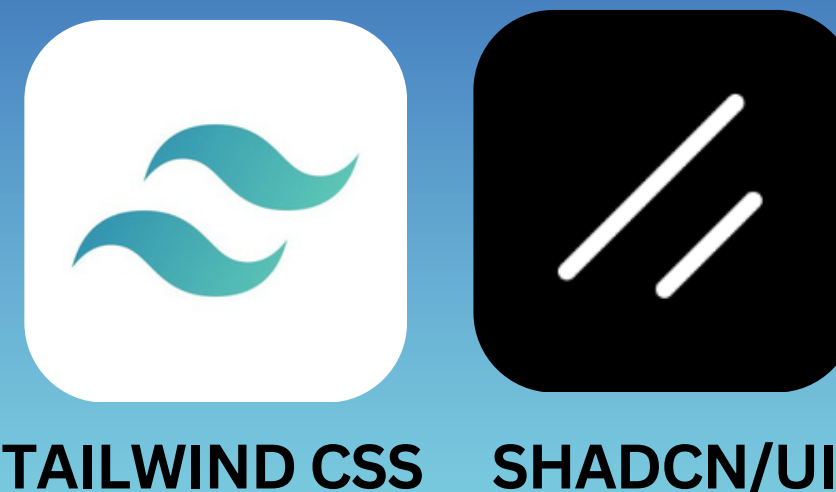
DATABASE



AI ANALYSIS



UI/UX



PROTOTYPE

PATIENT FORM

Pre-Consultation Form

AI Assistant - Fill Form by Voice

Or fill the form manually below

Patient Name: Ahmed Al-Mansour

National ID: 1345678901

Select Doctor: Faisal alnasyan

Gender: Male

Age: 58

Chronic Diseases: Diabetes, Hypertension

Describe Your Symptoms: I've been feeling very tired for the past week and my feet are swelling, especially in the evenings. Also having shortness of breath when I walk upstairs. I've been taking my diabetes and blood pressure medications regularly.

Submit Report

PATIENT REPORT

Patient Information

Name: Ahmed Al-Mansour, National ID: 1345678901, Age: 58 years, Gender: Male, Submitted On: May 5, 2026 at 12:10 AM

Chronic Diseases: Diabetes, Hypertension

Symptoms: I've been feeling very tired for the past week and my feet are swelling, especially in the evenings. Also having shortness of breath when I walk upstairs. I've been taking my diabetes and blood pressure medications regularly.

AI Analysis

DIFFERENTIAL DIAGNOSIS

1. Heart Failure - Most likely given peripheral edema, dyspnea on exertion, and fatigue. Patient's history of diabetes and hypertension are significant risk factors for cardiac dysfunction.

2. Diabetic Nephropathy - Consider renal impairment given chronic diabetes and hypertension. Lower extremity edema may indicate fluid retention from decreased kidney function.

3. Acute Coronary Syndrome - Must rule out given cardiovascular risk factors and symptom severity.

RED FLAGS

- Shortness of breath with exertion in high-risk patient warrants urgent cardiac evaluation

INITIAL WORKUP / ACTIONS

- Echocardiogram and ECG to assess cardiac function

- BNP levels and comprehensive metabolic panel

- Renal function tests (creatinine, eGFR)

- Consider cardiology referral for risk stratification

Report Status: New

Mark as Reviewed

PROJECT IMPACT / KEY BENEFITS



FOR PATIENTS

Faster Access: Submit consultation requests anytime, anywhere without waiting in lines.

Voice Convenience: Fill forms by speaking naturally in English or Arabic.

Better Preparation: Patients can thoughtfully describe symptoms before seeing the doctor.



FOR DOCTOR

Time Efficiency: Review patient information and AI analysis before appointments.

Informed Decisions: Preliminary diagnosis suggestions help focus the consultation.

Reduced Workload: Automated data collection eliminates manual paperwork

CONCLUSION

The system successfully demonstrates how AI can enhance medical pre-consultation workflows. The integration of voice-based interaction significantly improved accessibility, while automated analysis provided valuable preliminary insights to physicians. The bilingual support ensures wider applicability in diverse healthcare settings.

